

Lesson 7.2: Imperfect information

Information dynamics

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Oil prospecting: imperfect information

- ▶ unlikely that test is 100% reliable

- ▶ observables:

p test *positive*

n test *negative*

- ▶ suppose test only 80% reliable:

$$P(p|o) = 0.8$$

$$P(n|\bar{o}) = 0.8$$

Question

Would you pay for *this* test?

Decision analysis

- ▶ joint probabilities:

	o	$\bar{o}$	
p	0.4	0.1	0.5
n	0.1	0.4	0.5
	0.5	0.5	

- ▶ decision table:

	0.4	0.1	0.1	0.4	
	p	p	n	n	
	o	$\bar{o}$	o	$\bar{o}$	EMV
E	-1	-1	-1	-1	-1.0
D	30	-10	30	-10	10.0
ED	-1	-1	30	-10	-1.5
DE	30	-10	-1	-1	10.5

Final conclusion

ignoring test's cost:

strategy	EMV
D	10.0
DE	10.5

including test's cost (\$3):

strategy	EMV
D	10.0
DE	7.5

Conclusion

Test is not worth its cost.

Optimal strategy

forgo test; drill